28/9/26

Second order ODEs – Initial Value Problems

Example 3.10

$$(t^2-3t)y'' + ty' - (t+3)y = 0 \qquad y(1)=2 \quad y'(1)=1$$

(t_0 to y) (t_0 to y')

Rewrite as

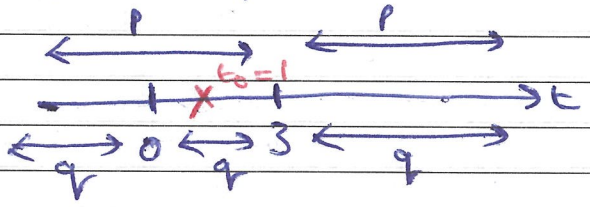
$$y'' + \frac{t}{t(t-3)}y' - \frac{(t+3)}{t(t-3)}y = 0 \qquad \text{for } t \neq 0, 3$$

$\therefore$ $y'' + py' + qy = g$ gives $p = \frac{1}{t-3}$ and $g = 0$

and $q = \frac{t+3}{t(t-3)}$

p is continuous for $t \in (-\infty, 3)$ or $t \in (3, \infty)$

q is continuous for $t \in (-\infty, 0)$, $(0, 3)$, $(3, \infty)$



The largest interval in which p, q, g are continuous and $t_0 \in I$ is $0 < t < 3$

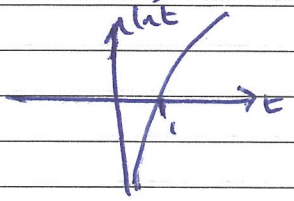
Example 3.11

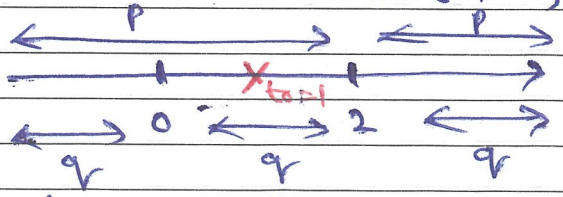
$$y'' + \frac{t}{t-2}y' + \frac{(\ln|t|)}{t-2}y = 0 \qquad y(1)=0 \quad y'(1)=1$$

(t_0 to y)

$p = \frac{t}{t-2}$ = cts for $t \in (-\infty, 2)$ and $t \in (2, \infty)$

$q = \frac{\ln|t|}{t-2}$ is cts for $t \in (-\infty, 0)$, $t \in (0, 2)$, $t \in (2, \infty)$





$\therefore$ The largest interval in which theorem 3.4 guarantees that there is a unique solution is $0 < t < 2$

Example 3.14

$$y'' + 5y' + 6y = 0$$

$y = e^{mt}$ $\quad m^2 + 5m + 6 = 0$

$(m+2)(m+3) = 0$

$\therefore m_1 = -3 \quad m_2 = -2$

General solution is $y = c_1 e^{-3t} + c_2 e^{-2t}$

Check (by substitution) that this is a solution.

Example 3.15

$$y'' + 5y' + 6y = 0 \quad y(0) = 2 \quad y'(0) = 3$$

From Ex 3.14 $y = c_1 e^{-3t} + c_2 e^{-2t}$

- $y(0) = 2 \Rightarrow 2 = c_1 e^{0} + c_2 e^{0} = c_1 + c_2$ (1)
- $y'(0) = 3 \Rightarrow y' = c_1(-3)e^{-3t} + c_2(-2)e^{-2t}$

$\Rightarrow y'(0) = -3c_1 e^{0} - 2c_2 e^{0} = 3$

$\Rightarrow -3c_1 - 2c_2 = 3$ (2)

Solve (1) and (2) for c_1 and c_2!

$c_1 = -7$ and $c_2 = 9$

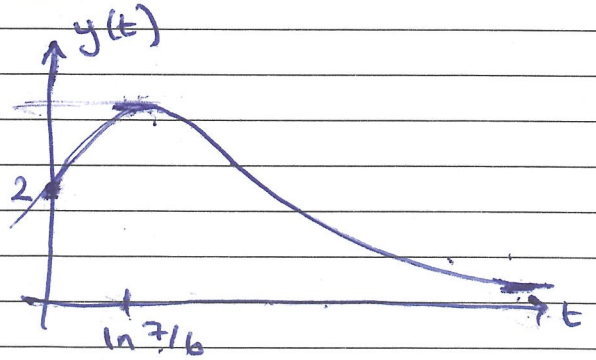
The solution of the IVP is

$$y = -7e^{-3t} + 9e^{-2t}$$

[Check!]

- $y(0) = 2$
- $y'(0) = 3$
- As $t \to \infty$ $y \to 0^+$
- $y = 0$ when $9e^{-2t} = 7e^{-3t}$

  Solve for t ...



- $\frac{dy}{dt} = 21e^{-3t} - 18e^{-2t} = 0$ when $t = \ldots$

The solution initially increases since $y'(0) > 0$ - at

The solution eventually approaches zero due to e

when $a > 0$ $\therefore$ $\exists$ maximum

Example 3.16

$$16y'' - 8y' + 145y = 0 \qquad y(0) = -2 \quad y'(0) = 1$$

Sub $y = e^{mt}$ $\quad 16m^2 - 8m + 145 = 0$

Solve for $m = \dfrac{8 \pm \sqrt{8^2 - 4 \cdot 16 \cdot 145}}{2 \cdot 16} = \dfrac{1}{4} \pm 3i = \lambda \pm i\mu$

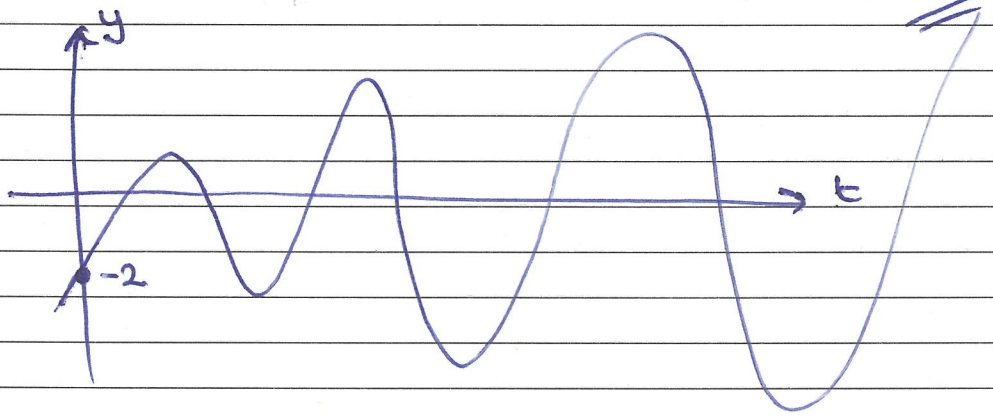
The general solution (3.17)

$$y = c_1 e^{t/4} \cos(3t) + c_2 e^{t/4} \sin(3t) \qquad t \in \mathbb{R}.$$

- $y(0) = c_1 e^{0} \cos(0) + c_2 e^{0} \sin(0) = c_1 = -2$
- $y'(0) = 1 \quad \Rightarrow \ldots \quad c_2 = 1/2$

Hence the solution to the IVP is

$$y = -2e^{t/4} \cos(3t) + \frac{1}{2} e^{t/4} \sin(3t) \qquad t \in \mathbb{R}.$$



- $y(0) = -2$
- $y'(0) = 1 > 0$
- oscillations
- $e^{t/4}$ growing amplitude

The solution grows as it oscillates

[$\lambda < 0$ decaying oscillation
$\lambda = 0$ fixed amplitude oscillation]